

JSON Schema for Prospectus/KID Data Extraction

We define a JSON schema capturing each column's meaning and extraction rules. Each field corresponds to information on the prospectus (e.g. Key Information Document or KID) linked by the **Link** URL. We use domain conventions (e.g. ISO date formats, numeric percentages) and cite examples from a sample SG KID

1 2 .

Identification Fields

Identification fields come from the document header or cover. For example, the SG KID shows the product name, ISIN, and manufacturer on page 1 3 , and the issuer/guarantor and KID ID on page 1 4 5 . We include:

- **number** – an integer index (row number, not from the document).
- **link** – the document URL (type: URI).
- **global_identifier** – the unique product/document code (e.g. the KID ID string “KID-AST-...” printed at the end of the document 5).
- **date_filed** – the document's production/filing date, as given on the KID (e.g. “Date of Production of this KID: 22/04/2024” 4). Format it as a date (YYYY-MM-DD or DD/MM/YYYY).
- **cusip** – the CUSIP security ID (9-character US/Canada code 6), if listed in the document.
- **isin** – the ISIN (12-character international security ID 7) from the header (e.g. “ISIN: DE000SR7YGD4” 3).
- **issuer** – the name of the issuing entity (e.g. “SG Issuer” 4).
- **guarantor** – the name of the guarantor if any (often the same as manufacturer, e.g. “Société Générale” 4).
- **distributor** – name of the selling agent or distributor, if present in the document.

```
{
  "number": {
    "type": "integer",
    "description": "Sequential row index (not in document).",
  },
  "link": {
    "type": "string",
    "format": "uri",
    "description": "URL of the prospectus/KID document.",
  },
  "global_identifier": {
    "type": "string",
    "description": "Unique product/document code, e.g. the KID ID (see “KID_ID” label in the KID) 5 .",
  },
  "date_filed": {
```

```

    "type": "string",
    "format": "date",
    "description": "Document production/filing date (as shown, e.g., "Date of
Production of this KID" 4 )."
  },
  "cusip": {
    "type": "string",
    "description": "CUSIP identifier (9-character security code)
6 , if provided in the document."
  },
  "isin": {
    "type": "string",
    "description": "ISIN code (12-character security code) 7 from the document
header (e.g. "ISIN: DE000SR7YGD4" 3 )."
  },
  "issuer": {
    "type": "string",
    "description": "Name of issuing entity (e.g. 'SG Issuer' as shown on the
KID) 4 ."
  },
  "guarantor": {
    "type": "string",
    "description": "Name of guarantor, if any (e.g. 'Société Générale' 4 )."
  },
  "distributor": {
    "type": "string",
    "description": "Name of the distributor or selling agent, if indicated."
  }
}

```

Date and Timing Fields

These fields capture key dates from the document's calendar or terms. For example, the KID's "Calendar" section shows **Issue Date**, **Optional Redemption Dates**, etc. ⁸ . We include:

- `issue_date` – the issue date of the product (from "Issue Date" ⁸ , e.g. "08/12/2020"). Use ISO date format.
- `trade_date` – date the product first trades (if specified).
- `offering_opens` / `offering_closes` – public offering start/end dates (if given).
- `listing_date` – the date of first listing on an exchange.
- `effective_date` – the effective date of the prospectus (or approval date).
- `maturity_date` – product maturity date (if it has one; in our example KID it is "no Maturity" ²).
- `expiry_date` – expiry or final settlement date for embedded options.
- `final_valuation_date` – the final valuation/fixing date (e.g. for barrier or range accrual products).

```

{
  "issue_date": {

```

```

    "type": "string",
    "format": "date",
    "description": "Issue date of the product (e.g. '08/12/2020' from the
document) 8 ."
  },
  "trade_date": {
    "type": "string",
    "format": "date",
    "description": "Trade date or first listing date, if specified."
  },
  "offering_opens": {
    "type": "string",
    "format": "date",
    "description": "Start date of the public offering (if given)."
  },
  "offering_closes": {
    "type": "string",
    "format": "date",
    "description": "End date of the public offering."
  },
  "listing_date": {
    "type": "string",
    "format": "date",
    "description": "Date of exchange listing."
  },
  "effective_date": {
    "type": "string",
    "format": "date",
    "description": "Effective date of the prospectus/registration (as stated in
document)."
  },
  "maturity_date": {
    "type": "string",
    "format": "date",
    "description": "Maturity date of the product (if any)."
  },
  "expiry_date": {
    "type": "string",
    "format": "date",
    "description": "Expiry or final settlement date for options (if
applicable)."
  },
  "final_valuation_date": {
    "type": "string",
    "format": "date",
    "description": "Last valuation/fixing date for payoff (if applicable)."
  }
}

```

Financial Terms

Numeric and monetary fields define the investment terms. In the KID example, the “Fix amount” and “Issue Price” appear under *Product* ². These include:

- `fix_amount` – the face value per unit (e.g. 100 for “EUR 100” ²).
- `issue_price` – the issue price, often given as a percentage of face value (e.g. “100%” of fix amount).
- `principal_amount` – nominal principal amount (if different from fix amount).
- `aggregate_principal_amount_type` – indicator if aggregate principal is fixed, notional, etc.
- `proportional_principal_amount` – share of principal if product is a fraction of a larger notional.
- `minimum_investment` – minimum subscription amount (e.g. 100 in ²).
- `maximum_investment` – maximum subscription, if any.
- `offering_size` – total size of the offering (in units or currency).
- `payment_currency` – currency of payoffs (e.g. “EUR” as in document header ²).
- `settlement_currency` – settlement currency (often same as payment currency).

```
{
  "fix_amount": {
    "type": "number",
    "description": "Face value per certificate (e.g. 100 for 'EUR 100' 2 )."
  },
  "issue_price": {
    "type": "string",
    "description": "Issue price as a percentage of face value (e.g. '100%')."
  },
  "principal_amount": {
    "type": "number",
    "description": "Nominal principal amount per unit, if different from fix
amount."
  },
  "aggregate_principal_amount_type": {
    "type": "string",
    "description": "Type of aggregate principal (e.g. 'Fixed' or 'Variable')."
  },
  "proportional_principal_amount": {
    "type": "number",
    "description": "Proportional share of principal in case of split/
participation."
  },
  "minimum_investment": {
    "type": "number",
    "description": "Minimum investment amount (e.g. 100 as in 2 )."
  },
  "maximum_investment": {
    "type": "number",
    "description": "Maximum investment amount allowed."
  },
}
```

```

"offering_size": {
  "type": "number",
  "description": "Total offering size (number of units or total value).",
},
"payment_currency": {
  "type": "string",
  "description": "Currency of payoffs (e.g. 'EUR').",
},
"settlement_currency": {
  "type": "string",
  "description": "Settlement currency (usually same as payment currency).",
}
}

```

Product Classification

These fields categorize the product. From [22], *Type* is given as “certificate” ⁹, and underlyings are listed (e.g. index ISIN DE000SL0BE56 ¹⁰). Fields include:

- `document_type` – type of document (“KID”, “prospectus”, etc.).
- `product_type` – general type (e.g. “certificate” ⁹, “bond”, “option”).
- `product_subtype` – more specific (e.g. “Equity Linked Certificate”).
- `fixed_income_type` – if fixed income, the category (e.g. “Bond” or “Note”).
- `investment_type` – product objective category (e.g. “Capital Growth”, “Yield”).
- `underlying_ticker` – identifier (ticker or ISIN) of the underlying asset (e.g. index ISIN ¹⁰).
- `multiple_underlyings` – true if a basket of assets.
- `basket_weightings_unequal` – true if basket constituents have unequal weights.

```

{
  "document_type": {
    "type": "string",
    "description": "Type of document (e.g. 'KID' or 'Prospectus').",
  },
  "product_type": {
    "type": "string",
    "description": "General product category (e.g. 'certificate' 9, 'bond',
etc.).",
  },
  "product_subtype": {
    "type": "string",
    "description": "Specific product category (e.g. 'Equity Linked
Certificate').",
  },
  "fixed_income_type": {
    "type": "string",
    "description": "Type if fixed-income product (e.g. 'Bond', 'Note').",
  },
}

```

```

"investment_type": {
  "type": "string",
  "description": "Investment objective/category (e.g. 'Capital Growth')."
},
"underlying_ticker": {
  "type": "string",
  "description": "Identifier of underlying asset (ticker or ISIN 10)."
},
"multiple_underlyings": {
  "type": "boolean",
  "description": "True if the product has multiple underlying assets (basket)."
},
"basket_weightings_unequal": {
  "type": "boolean",
  "description": "True if basket underlyings have unequal weights."
}
}

```

Coupon and Payoff Features

Structured products may pay coupons or have barrier/digital features. These fields come from sections like “Coupon” or “Redemption”. For example, if a product has a fixed coupon, the KID would specify a rate. We capture:

- `coupon_applies` – true if a coupon exists.
- `coupon_rate` – fixed coupon rate (%) per period.
- `coupon_interval` – frequency of coupon payments (e.g. “Quarterly”).
- `coupon_schedule` – details of coupon payment dates.
- `coupon_first_date` / `coupon_last_date` – first and final coupon dates.
- `contingent_coupon_applies` – true if coupon is contingent on an underlying condition.
- **Barrier Levels:** e.g. `knock_in_lower_barrier_level`, `knock_out_upper_barrier_level` – numeric barrier levels (as % of initial underlying).
- `knock_out_upper_barrier_event_types` – description of events triggering knock-out (e.g. “on each observation date”).
- Similarly for lower knock-out (if any).
- `digital_strike_level`, `put_strike_level`, `call_strike_level` – strike prices for embedded digital, put or call options (in % or absolute terms).
- `buffer_level` – protection/buffer level if applicable.

```

{
  "coupon_applies": {
    "type": "boolean",
    "description": "Whether the product pays a periodic coupon."
  },
  "coupon_rate": {

```

```

    "type": "number",
    "description": "Fixed coupon rate (%) per payment period."
  },
  "coupon_interval": {
    "type": "string",
    "description": "Coupon payment frequency (e.g. 'Quarterly')."
  },
  "coupon_schedule": {
    "type": "string",
    "description": "Schedule or dates of coupon payments."
  },
  "coupon_first_date": {
    "type": "string",
    "format": "date",
    "description": "First coupon payment date."
  },
  "coupon_last_date": {
    "type": "string",
    "format": "date",
    "description": "Final coupon payment date."
  },
  "contingent_coupon_applies": {
    "type": "boolean",
    "description": "True if the coupon depends on underlying performance."
  },
  "knock_in_lower_barrier_level": {
    "type": "number",
    "description": "Lower knock-in barrier level (% of initial underlying)."
  },
  "knock_out_upper_barrier_level": {
    "type": "number",
    "description": "Upper knock-out barrier level (%)."
  },
  "knock_out_upper_barrier_event_types": {
    "type": "string",
    "description": "Conditions that trigger a knock-out (e.g. observation
frequency)."
  },
  "knock_out_lower_barrier_level": {
    "type": "number",
    "description": "Lower knock-out barrier level (%)."
  },
  "knock_out_lower_barrier_event_types": {
    "type": "string",
    "description": "Conditions triggering a lower knock-out event."
  },
  "digital_strike_level": {
    "type": "number",

```

```

    "description": "Strike level for a digital (binary) payoff."
  },
  "put_strike_level": {
    "type": "number",
    "description": "Strike level for an embedded put option."
  },
  "call_strike_level": {
    "type": "number",
    "description": "Strike level for an embedded call option."
  },
  "buffer_level": {
    "type": "number",
    "description": "Buffer protection level (%) (for buffer products).".
  }
}

```

Autocall/Callable Features

If the product can be called early (autocallable), it will have call schedule and barriers. Typical fields:

- `autocall_applies` - true if autocall feature is present.
- `autocall_first_date` / `autocall_last_date` - observation window for autocall.
- `autocall_barrier` - barrier level (%) for autocall redemption.
- `autocall_first_payment_date` / `autocall_last_payment_date` - payment dates if autocall triggers.
- (Similarly, `issuer_call_*` fields for issuer-exercise features.)

```

{
  "autocall_applies": {
    "type": "boolean",
    "description": "True if the product has an autocallable early redemption feature."
  },
  "autocall_first_date": {
    "type": "string",
    "format": "date",
    "description": "First possible autocall observation date."
  },
  "autocall_last_date": {
    "type": "string",
    "format": "date",
    "description": "Last possible autocall observation date."
  },
  "autocall_barrier": {
    "type": "number",
    "description": "Barrier level (%) for autocall redemption."
  },
}

```

```

"autocall_first_payment_date": {
  "type": "string",
  "format": "date",
  "description": "Payment date if autocall triggers at first opportunity."
},
"autocall_last_payment_date": {
  "type": "string",
  "format": "date",
  "description": "Payment date if autocall triggers at last opportunity."
}
}

```

Credit-Linked and Other Features

For credit-linked notes (CLNs), fields prefixed `cln_` capture credit event details. E.g., `cln_recovery_rate` (%) and `cln_credit_events` (type of default events) are found in the credit section. Similarly, digital-conversion (`dcc_*`) or multi-range (`multi_range_*`) fields cover complex payoffs (these appear in specialized product descriptions). For brevity, we summarize categories:

- **Credit-Linked Note (CLN):**

- `cln_recovery_rate` (number) – recovery rate (%) after credit event.
- `cln_credit_events` (string) – description of credit events.
- `cln_reference_obligation_isin` (string) – ISIN of reference bond.

- (And other `cln_` fields as needed, e.g. observation dates.)

- **Multi-Range & Range-Accrual:**

- Fields like `range_accrual_barrier_level_absolute`, `range_accrual_upper_barrier_level_absolute`, etc., define barrier levels for range accrual products.
- `multi_range_barrier_level_1`, `multi_range_results_1`, etc., capture piecewise payoff ranges (see structured payoff table).

- **Digital Conversion (DCC):**

- Fields such as `dcc_base_currency`, `dcc_reference_currency`, `dcc_conversion_event_type`, `dcc_strike_rate`, etc., describe digital conversion conditions.

These fields are extracted from the sections detailing those features (often tables or bullet points in the document). Each is typed appropriately (dates in ISO format, numeric percentages, or boolean flags).

```

{
  "cln_recovery_rate": {
    "type": "number",

```

```

    "description": "Recovery rate (%) in case of a credit event (for credit-linked notes).",
  },
  "cln_credit_events": {
    "type": "string",
    "description": "Description of credit events triggering payout adjustments."
  },
  "cln_reference_obligation_isin": {
    "type": "string",
    "description": "ISIN of the reference obligation/bond for a CLN."
  }
  /* Additional cln_* fields as needed */
}

```

```

{
  "range_accrual_barrier_level_absolute": {
    "type": "number",
    "description": "Barrier level (%) for range-accrual features (absolute terms).",
  },
  "range_accrual_upper_barrier_level_absolute": {
    "type": "number",
    "description": "Upper barrier level (%) for range accrual."
  },
  "range_accrual_lower_barrier_level_absolute": {
    "type": "number",
    "description": "Lower barrier level (%) for range accrual."
  }
  /* Additional multi_range_* fields for multi-range payoffs */
}

```

Data Formats and Conditions

- **Dates:** Parse dates (often in DD/MM/YYYY) to ISO format (YYYY-MM-DD). If a date field is blank or “N/A”, set it null.
- **Numeric values:** Strip symbols (“%”, “EUR”) and convert to numbers. E.g. “100%” becomes 100 (or 1.0 as needed), “EUR 100” becomes 100.00.
- **Booleans:** Use true/false for apply-fields (e.g. `coupon_applies`). If a feature is absent in the document, these flags are false.
- **Optional fields:** Many fields apply only to certain products. The schema allows them but they can be null or omitted if not relevant. For example, a pure certificate has no coupon, so `coupon_applies=false` and `coupon_rate` is null.

By combining this schema with NLP extraction, an LLM can identify each item in the document (e.g. by key phrases like “ISIN”, “Issue Date”, “Fix amount”, etc.) and fill the corresponding JSON fields. Domain rules (e.g.

“Issue Price is a percentage of Fix Amount”, “No maturity means null date”) should be applied during parsing.

Sources: We base these definitions on standard KID/prospectus structure and identifiers ¹ ². In particular, product ISIN and issuer appear at the top ¹, and issue terms (fix amount, price, dates) appear in the product section ² ⁸. (ISIN and CUSIP definitions from finance references ⁷ ⁶.)

¹ ² ³ ⁴ ⁵ ⁸ ⁹ ¹⁰ KID-AST-z7GrZZ55n-eng

<https://kid.sgmarkets.com/kid-id/KID-AST-z7GrZZ55n/eng>

⁶ CUSIP Explained: The Key to Efficient Securities Identification and Tracking

<https://cbonds.com/glossary/cusip/>

⁷ Understanding ISIN: Definition, Uses, and Importance in Global Finance

<https://www.investopedia.com/terms/i/isin.asp>